

F78PB Probability and Statistics B

Constructing Estimators

Construction and estimation of statistical models

We now consider the problem of constructing an estimator for a parameter θ. We will consider two approaches:

1. Method of Moments estimation
2. Maximum likelihood estimation

Method of Moments estimation (MME)

Consider an experiment that yields an i.i.d. set of observations $X_1, ..., X_n$ from a distribution with density or mass function $f_X(x; \theta)$, where θ is unknown parameter(s).

Sometimes θ will be a *vector* of parameters,
e.g. if $X \sim \text{Gamma}(\alpha, \beta)$ then $\theta = (\alpha, \beta)$ (2-dimensional vector).

To find the MME for θ we

i. Equate the sample moments with theoretical (population) moments.
ii. Then solve the equation(s).

Example (Exit poll)

Recall the results of an *exit poll* give a random sample $X_1, \ldots, X_m$ from a Ber(p) distribution (where voting for candidate A is signified by $X_i = 1$). A natural estimator is $\hat{p}(\underline{X}) = \frac{1}{m}\sum X_i$.

How could we have derived this systematically using MME?

Solution:
For the Ber(p) distribution, the first moment is:

$$\mathbb{E}(X) = \mu(p) = p$$

as a function of p, $\mu(p)$. We equate this 'population' moment to the sample mean,

$$\bar{X} = \frac{\sum X_i}{m}$$

and solve for the parameter p to obtain the estimator

$$\mathbb{E}(X) = \bar{X} \Rightarrow \hat{p}(\underline{X}) = \bar{X}.$$

Method of Moments for 1 parameter

The method of moments estimator for a 1-dimensional parameter θ is found by solving the equation

$$\mu(\theta) = \bar{X}.$$

Example

Suppose $X_1, ..., X_n$ is a random sample from Exp(β). What is the MME of β?

Solution: ...

Example

Now let $X_1, ..., X_n$ be a random sample from Gamma (α, β). What is the MME for $\theta = (\alpha, \beta)$?

Solution: ...

Method of Moments for k parameters

We can perform method-of-moments estimation for distributions that have k unknown parameters: $\theta = (\theta_1, \theta_2, \ldots, \theta_k)$:

- Consider all moments up to order k and equate the rth sample moment, $\nu_r = \frac{1}{n}\sum x_i^r$, with the rth population moment, $\mathbb{E}[X^r]$, which is a function of the parameter θ, i.e. $\mathbb{E}[X^r] = \mu_r(\theta)$.
- This generates k simultaneous equations of the form

$$\nu_r = \mu_r(\theta), r = 1, \ldots, k$$

  which we then solve for θ to get the estimators.

Note that equivalently we could work with central moments $\mathbb{E}(X - \mu)^r$ equated with their sample values $m_r = \frac{1}{n}\sum(x_i - \bar{x})^r$.

Advantages and disadvantages of MME

- Main advantage is that it is generally simple to apply
- However, it has the disadvantage that in some circumstances it can give results that are clearly nonsensical.

Example

Suppose that you conduct an experiment in which you observe $X_1, \ldots, X_n$ where the X_i are i.i.d. from a $\mathrm{U}(0, \theta)$, where $\theta > 0$. Find the MME of θ.

Solution:
We know that $\mathbb{E}(X) = \frac{\theta}{2}$, so

$$\bar{X} = \frac{\theta}{2} \Rightarrow \hat{\theta}(\underline{X}) = 2\bar{X}.$$

Now suppose that for the particular experiment $n = 3$ and $(x_1, x_2, x_3) = (0.1, 0.2, 0.9)$.
Then in this case we would obtain $\hat{\theta}(\underline{x}) = 2\bar{x} = 0.8$.
However, this doesn't make sense: the data are not consistent with this value of θ! Why?

Maximum Likelihood Estimation

Idea

Find the parameter value θ for which the observed data are most likely.

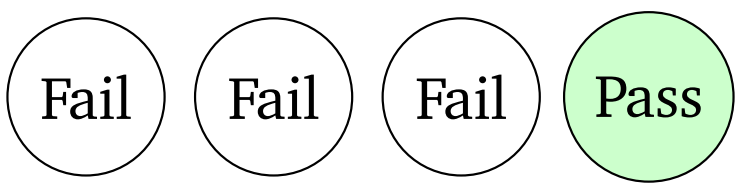
Example:

- A driver passes the test on the 4th attempt.
- Goal: Estimate probability θ of passing on any attempt.

Assumptions

- Attempts are independent.
- Probability θ is constant across attempts.

Visual representation:



Observed data: $\underline{x} = (0, 0, 0, 1)$

Choosing θ

Observed data: $\underline{x} = (0, 0, 0, 1)$

Probability of observing this data:

$$\mathbb{P}(\underline{X} = \underline{x}; \theta) = \mathbb{P}(X_1 = 0, X_2 = 0, X_3 = 0, X_4 = 1; \theta) = (1 - \theta)^3 \theta$$

Compute likelihood for each candidate:

θ	$\mathbb{P}(\underline{X} = \underline{x}; \theta)$
0.25	0.105
0.50	0.063
0.75	0.012

Most likely when $\theta = 0.25$.

Likelihood

Definition

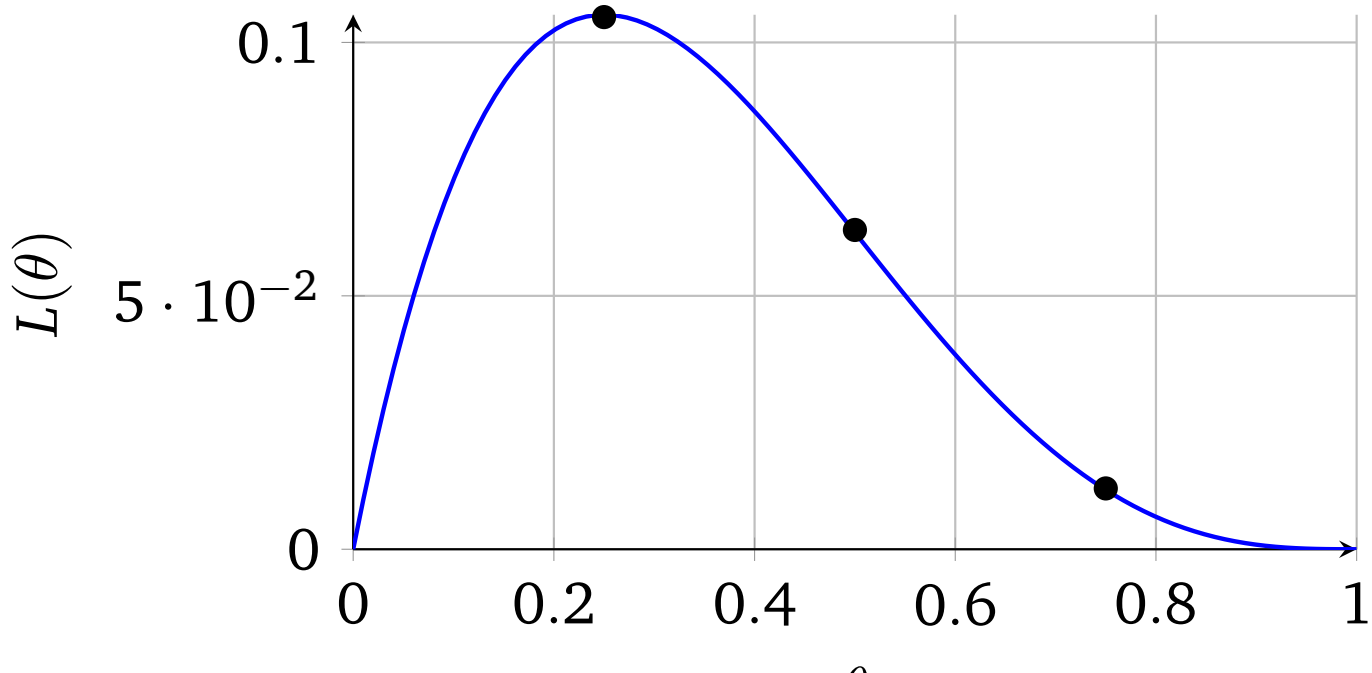
Likelihood function:

$$L(\theta; \underline{y}) = \mathbb{P}(\underline{Y} = \underline{y}; \theta)$$

It tells us how likely the observed data are for a given θ.

Note: We study how $L(\theta; \underline{y})$ varies with θ; the data $\underline{y}$ are fixed.

Visual: Likelihood curve for $(1-\theta)^3\theta$



Likelihood for i.i.d. observations from a discrete distribution

Let $Y_1, \ldots, Y_n$ be i.i.d. variables from a discrete distribution with probability mass function $p_Y(y;\theta)$.

Suppose the *realised* values from the experiment are $\underline{y} = (y_1, \ldots y_n)$.

Then the likelihood function is given by

$$\begin{aligned} L(\theta;\underline{y}) &= \mathbb{P}(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n|\theta) \\ &= p_Y(y_1;\theta) \times p_Y(y_2;\theta) \times \cdots \times p_Y(y_n;\theta) \\ &\overset{\text{independence}}{=} \prod_{i=1}^{n} p_Y(y_i;\theta) \end{aligned}$$

We then obtain the MLE by maximising $L(\theta;\underline{y})$ with respect to θ (using calculus):

$$\hat{\theta}(\underline{y}) = \arg\max_{\theta} L(\theta;\underline{y}).$$

(Here, $\arg\max$ *means "the value of* θ *that gives the largest likelihood.")*

Often it is more convenient to maximise the logarithm of the likelihood

$$\ell(\theta;\underline{y}) = \log(L(\theta;\underline{y})) = \sum_{i=1}^{n} \log(p_Y(y_i;\theta)).$$

Note that since $\log$ is a monotonic increasing function, the value of θ that maximises $\ell(\theta)$ also maximises $L(\theta)$ and *vice versa*.

Example

An astronomer counts the number of meteors in a section of the night sky for n consecutive nights obtaining data $x_1, \ldots, x_n$. She believes that these values are a random sample (i.i.d.) from a Poi(λ) distribution where λ is unknown. What is the MLE of λ?

Solution:
Since the observations are assumed to be i.i.d. we have

$$L(\lambda; \underline{x}) = \prod_{i=1}^{n} p_X(x_i; \lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = \frac{e^{-n\lambda}\lambda^{\sum x_i}}{\prod x_i!}$$

Taking the log, differentiating, equating to zero and solving for λ, we obtain

$$\hat{\lambda}(\underline{x}) = \frac{\sum x_i}{n} = \bar{x} \quad [\textit{exercise}]$$

We see that the MLE of λ is the same as we would obtain using method of moments. *This is not always the case!*

Example (Exit poll cont)

Suppose that the results of the *exit poll* gave that r students (out of m) voted for candidate A (and $m - r$ didn't). Find the MLE for p.

Solution: ...

Likelihood for i.i.d. Continuous Observations

Setup:

- Observations: $X_1, \ldots, X_n$ i.i.d. from a continuous distribution.
- Density: $f_X(x;\theta)$.

Problem

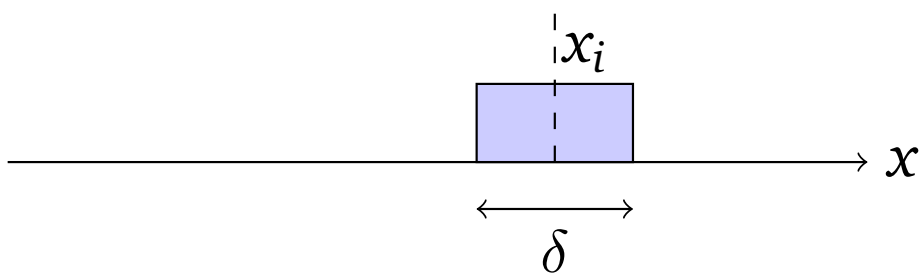
For continuous data:

$$\mathbb{P}(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n; \theta) = 0 \quad \text{(any } \theta\text{)}$$

Why? Exact values have zero probability in a continuous distribution.

Intuition: Measurements have finite precision (e.g., $\pm\delta/2$).

Visual:



Constructing the Likelihood

Idea: Approximate probabilities using intervals of width δ.

For one observation

$$\mathbb{P}\left(X_i \in \left(x_i - \frac{\delta}{2}, x_i + \frac{\delta}{2}\right)\right) = \int_{x_i-\delta/2}^{x_i+\delta/2} f_x(y)\, dy \approx f_X(x_i;\theta)\,\delta$$

Using this approximation and independence:

$$\begin{aligned}
&\mathbb{P}\left(X_1 \in \left(x_1 - \frac{\delta}{2}, x_1 + \frac{\delta}{2}\right), \ldots, X_n \in \left(x_n - \frac{\delta}{2}, x_n + \frac{\delta}{2}\right)\right) \\
&= \mathbb{P}\left(X_1 \in \left(x_1 - \frac{\delta}{2}, x_1 + \frac{\delta}{2}\right)\right) \times \ldots \times \mathbb{P}\left(X_n \in \left(x_n - \frac{\delta}{2}, x_n + \frac{\delta}{2}\right)\right) \\
&\approx \prod_{i=1}^{n} f_X(x_i;\theta)\,\delta^n
\end{aligned}$$

Constructing the Likelihood

$$\mathbb{P}\left(X_1 \in \left(x_1 - \frac{\delta}{2}, x_1 + \frac{\delta}{2}\right), \ldots, X_n \in \left(x_n - \frac{\delta}{2}, x_n + \frac{\delta}{2}\right)\right) \approx \prod_{i=1}^{n} f_X(x_i;\theta)\,\delta^n$$

Likelihood for n observations:

$$L(\theta;\underline{x}) = \prod_{i=1}^{n} f_X(x_i;\theta)$$

The MLE is the value of θ that maximises $L(\theta;\underline{x})$.

Example

Suppose $X_1, \ldots, X_n$ is a random sample from Exp(β). Find the MLE of β.

Solution:
The probability density function is $f_X(x; \lambda) = \lambda e^{-\lambda x}$.

For given $\underline{x} = (x_1, \ldots, x_n)$, the likelihood is

$$L(\lambda; \underline{x}) = \prod_{i=1}^{n} f_X(x_i; \lambda) = \lambda^n e^{-\lambda \sum x_i}$$

Taking logs and maximising you should find that ML estimate is

$$\hat{\lambda}(\underline{x}) = \frac{1}{\bar{x}}.$$

How does this estimate compare with the method-of-moments estimate?

Calculus will not always help to maximise the likelihood. Consider the following example.

Example

Let $X_1, \ldots, X_n$ be a random sample from a $U(0, \theta)$ distribution, where $\theta > 0$. What is the MLE of θ?

Solution:
We know that

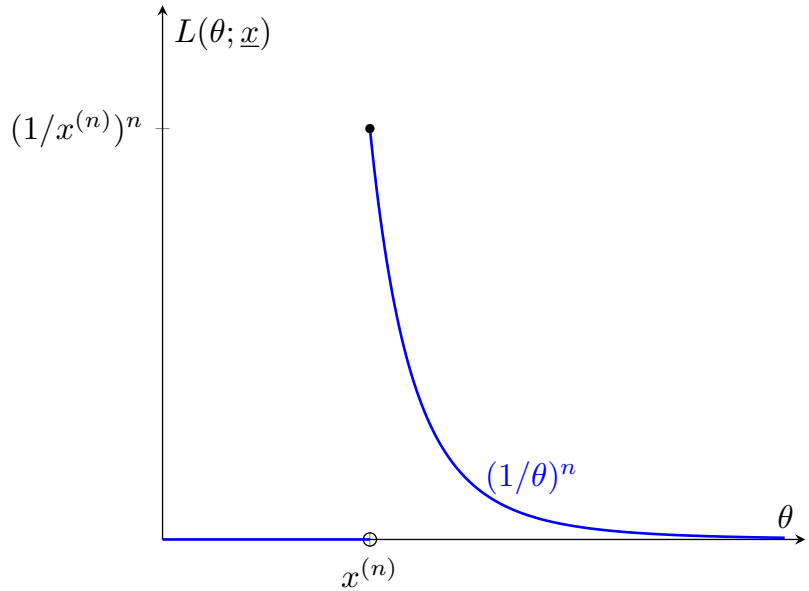
$$f(x;\theta) = \begin{cases} \frac{1}{\theta}, & \text{if } 0 < x < \theta \\ 0, & \text{otherwise} \end{cases}$$

Now the likelihood function is given by

$$L(\theta;\underline{x}) = \prod_{i=1}^{n} f(x_i;\theta) = \begin{cases} \left(\frac{1}{\theta}\right)^n, & \text{if } \theta \geq x_{(n)} = \max\{x_1, \ldots, x_n\} \\ 0, & \text{otherwise} \end{cases}$$

We cannot use calculus here - it is not clear which value of θ maximises $L(\theta;\underline{x})$.

Instead we consider the *graph* of $L(\theta; x)$ as a function of θ:



$L(\theta; \underline{x})$ is maximised for smallest possible value of θ.

It follows that the maximum likelihood estimator of θ is

$$\hat{\theta}(\underline{x}) = x_{(n)} = \max\{x_1, x_2, \ldots, x_n\}.$$

(Note this is not the same as MME.)

Problem:

Find the Bias and MSE of the MLE for θ. Compare your results with the corresponding ones for the MME. Which estimator do you think is better?

MLEs for several parameters

Suppose $\underline{X}$ is a random sample from a population pdf/pmf $f_X(x;\underline{\theta})$ where $\underline{\theta} = (\theta_1, \ldots, \theta_k)$.

We can use MLE to find a *vector* of estimators $\hat{\underline{\theta}} = (\hat{\theta}_1, \ldots, \hat{\theta}_k)$ of $\underline{\theta}$.

1. Write down the likelihood function $L(\underline{\theta};\underline{x}) = \prod_{i=1}^{n} f_X(x_i;\underline{\theta})$
2. Consider the log-likelihood function $\ell(\underline{\theta};\underline{x}) = \log L(\underline{\theta};\underline{x})$ and solve (*if possible*) the system of k equations

$$\frac{\partial}{\partial \theta_r}\ell(\underline{\theta};\underline{x}) = 0, \quad r = 1, 2, \ldots, k$$

   simultaneously to obtain $\hat{\underline{\theta}} = (\hat{\theta}_1, \ldots, \hat{\theta}_k)$.
3. Check whether $L(\underline{\theta};\underline{x})$ is maximised at $\hat{\underline{\theta}}$ (solution in step (2)). For $k = 2$ the 2-dimensional 2nd derivative test can be used.

Second Derivative Test in Two Variables

Setup: Let $f : \mathbb{R}^2 \to \mathbb{R}$ be twice continuously differentiable and (x^*, y^*) a critical point ($\nabla f(x^*, y^*) = 0$). The Hessian at (x^*, y^*) is:

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial x \partial y} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix}.$$

Compute:

$$D = \det(H) = \frac{\partial^2 f}{\partial x^2} \frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2.$$

Classification:

- $D > 0$, $\frac{\partial^2 f}{\partial x^2} > 0$: Local minimum.
- $D > 0$, $\frac{\partial^2 f}{\partial x^2} < 0$: Local maximum.
- $D < 0$: Saddle point.
- $D = 0$: Inconclusive.

Example

Suppose $X_1, \ldots, X_n$ is a random sample from $\mathrm{N}\left(\mu, \sigma^2\right)$. Find the MLE for μ and σ^2.

Solution:
The likelihood function is given by

$$L(\mu, \sigma^2; \underline{x}) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2\right)$$

$$\ell(\mu, \sigma^2; \underline{x}) = \log L(\mu, \sigma^2; \underline{x}) = -n\log(\sqrt{2\pi}) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Differentiate the log-likelihood function with respect to μ and σ^2 to get the following:

$$\frac{\partial}{\partial\mu}\ell(\mu, \sigma^2; \underline{x}) = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu)$$

$$\frac{\partial}{\partial\sigma^2}\ell(\mu, \sigma^2; \underline{x}) = \frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Solving $\frac{\partial \ell}{\partial \mu} = 0$ and $\frac{\partial \ell}{\partial \sigma^2} = 0$ corresponds to solving:

$$\frac{1}{\sigma^2} \sum_{i=1}^{n} (x_i - \mu) = 0 \tag{1}$$

$$\frac{-n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^{n} (x_i - \mu)^2 = 0. \tag{2}$$

$(1) \Rightarrow \hat{\mu}(\underline{x}) = \bar{x}$.

Next, substitute the estimate $\bar{x}$ for μ into the second equation, and solve for σ^2. We get

$$\hat{\sigma}^2(\underline{x}) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 = \frac{n-1}{n} s^2$$

Now, check if ℓ is *maximised* at $\hat{\mu} = \bar{x}$, $\hat{\sigma^2} = \frac{n-1}{n}s^2$.

Use the *2-dimensional* 2nd derivative test:

Note that:

$$\begin{aligned}
\frac{\partial^2}{\partial \mu^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) &= \frac{-n}{\hat{\sigma}^2} < 0 \\
\frac{\partial^2}{\partial (\sigma^2)^2}\ell(\hat{\mu}, \hat{\sigma^2}; \underline{x}) &= \frac{n}{2\hat{\sigma}^4} - \frac{1}{\hat{\sigma}^6}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n}{2\hat{\sigma}^4} - \frac{n\hat{\sigma}^2}{\hat{\sigma}^6} = \frac{-n}{2\hat{\sigma}^4} < 0 \\
\frac{\partial^2}{\partial \mu \partial \sigma^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) &= -\frac{1}{\hat{\sigma}^4}\sum_{i=1}^{n}(x_i - \bar{x}) = 0
\end{aligned}$$

Therefore,

$$\frac{\partial^2}{\partial \mu^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) \cdot \frac{\partial^2}{\partial (\sigma^2)^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x}) - \left(\frac{\partial^2}{\partial \mu \partial \sigma^2}\ell(\hat{\mu}, \hat{\sigma}^2; \underline{x})\right)^2 = \frac{n^2}{2\hat{\sigma}^6} > 0.$$

So, $L(\mu, \sigma^2; \underline{x})$ is *maximized* at $\hat{\mu}(\underline{x}) = \bar{x}, \hat{\sigma}^2(\underline{x}) = \frac{n-1}{n}s^2$, and this is a global maximum (since there is only one critical point).

Note that the MLE of σ^2 is *biased* (since $\mathbb{E}(S^2) = \sigma^2$).

Invariance of MLEs under One-to-One Transformations

Property: If two parameters are related by a one-to-one function

$$\varphi = h(\theta),$$

then their MLEs satisfy

$$\widehat{\varphi} = h(\widehat{\theta}).$$

Example: Let $X \sim \text{Exp}(\lambda)$, where the mean is

$$\mu = \mathbb{E}(X) = \frac{1}{\lambda}.$$

We know:

$$\widehat{\lambda}_{\text{MLE}}(\underline{x}) = \frac{1}{\bar{x}}.$$

Since $\mu = h(\lambda) = \frac{1}{\lambda}$, by invariance:

$$\widehat{\mu}_{\text{MLE}}(\underline{x}) = h\left(\widehat{\lambda}_{\text{MLE}}(\underline{x})\right) = \frac{1}{\widehat{\lambda}_{\text{MLE}}(\underline{x})} = \bar{x}.$$

Key takeaway: To find the MLE of a transformed parameter, apply the transformation to the original MLE.

Censored Observations

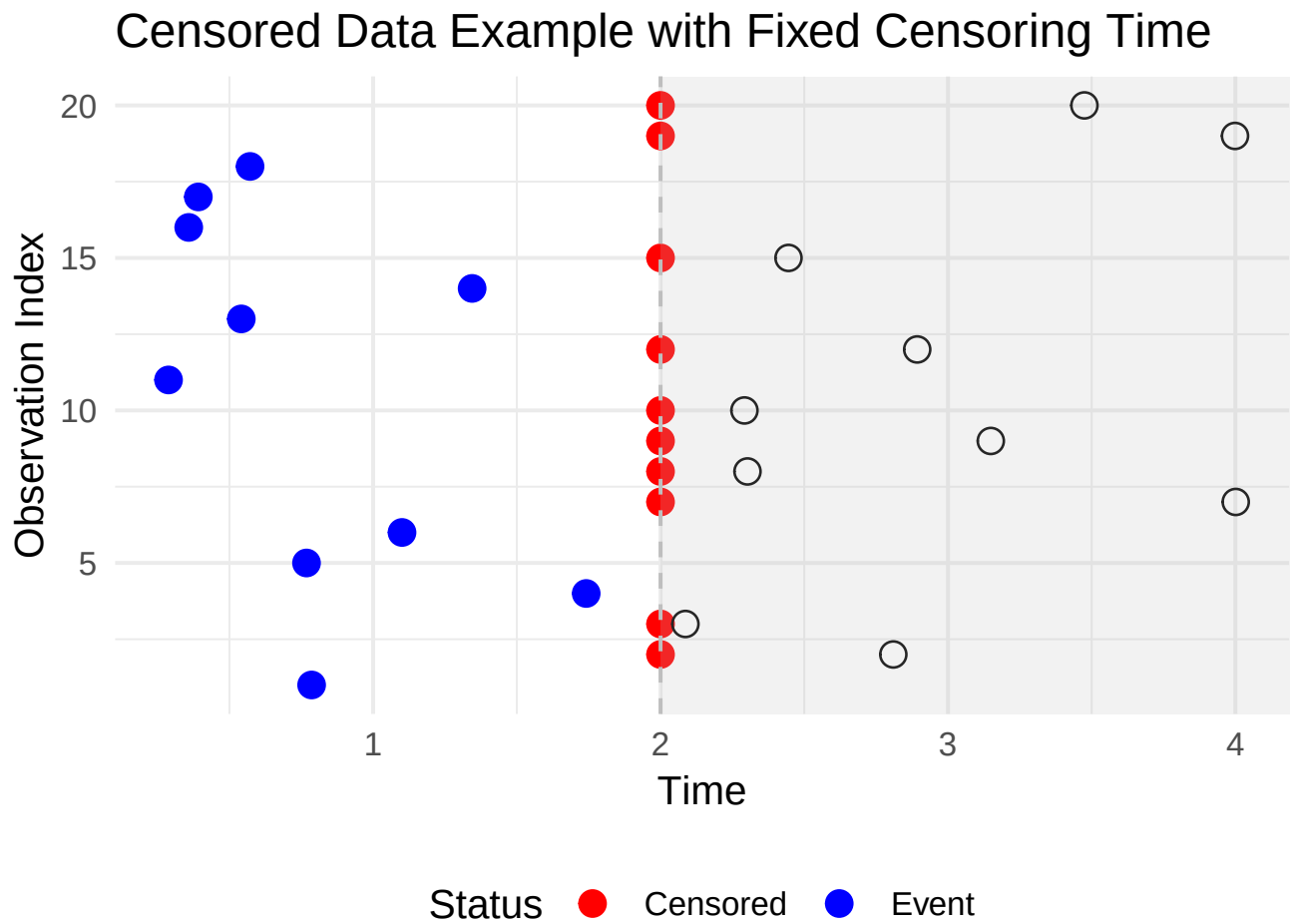
Definition: Observations that are only *partially known* are called *censored*.

Example: In a medical study:

- 20 patients followed for 2 months.
- Time T until condition develops, $T \sim \text{Exp}(\lambda)$.
- After 2 months:
  - 10 patients developed condition at times $t_1, \ldots, t_{10}$ with $\sum t_i = 7.3$.
  - 10 patients remained condition-free (censored at 2 months).

Goal: Construct likelihood and estimate λ.

Illustration



Likelihood Components

Assumption: Times are independent.

Observed patients: For $t_i < 2$ months, $1 \leq i \leq 10$:

$$f_T(t_i; \lambda) = \lambda e^{-\lambda t_i}$$

Censored patients: Probability that $T > 2$:

$$\mathbb{P}(T > 2) = e^{-2\lambda}$$

Constructing the Likelihood

Combine contributions:

$$L(\lambda; \underline{t}) = \prod_{i=1}^{10} \lambda e^{-\lambda t_i} \times \prod_{i=11}^{20} e^{-2\lambda}$$

Simplify:

$$L(\lambda; \underline{t}) = \lambda^{10} e^{-\lambda(20+\sum_{i=1}^{10} t_i)}$$

Log-likelihood:

$$\ell(\lambda) = 10 \log \lambda - \lambda(20 + \sum t_i)$$

Maximum Likelihood Estimate

Differentiate and set to zero (*exercise*):

$$\hat{\lambda} = \frac{10}{20 + \sum t_i}$$

Interpretation:

$$\hat{\lambda} = \frac{\text{Number of failures}}{\text{Total time at risk}}$$

This is a natural estimator of a *failure rate*.

Logistic Distribution

Example

Suppose that $\underline{X} = (X_1, \ldots, X_n)$ is a random sample from the logistic distribution with parameter θ. Note that the probability density function of the logistic distribution is

$$f(x;\theta) = \frac{\exp(-(x-\theta)}{(1+\exp(-(x-\theta)))^2}, \quad x \in \mathbb{R}.$$

Find the maximum likelihood estimate of θ.

The log-likelihood is

$$\ell(\theta;\underline{x}) = \sum_{i=1}^{n} \left[-(x_i-\theta) - 2\log(1+\exp(-(x_i-\theta)))\right].$$

The log-likelihood is

$$\ell(\theta;\underline{x}) = n\theta - n\bar{x} - 2\sum_{i=1}^{n} \log(1+\exp(-(x_i-\theta))).$$

The first derivative is

Logistic Distribution

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$$f(x;\theta) = \frac{\exp(-(x-\theta))}{(1+\exp(-(x-\theta)))^2}, \quad x \in \mathbb{R}.$$

Find the maximum likelihood estimate of θ.

The Maximum likelihood estimate is the solution of

$$\frac{n}{2} = \sum_{i=1}^{n} \frac{\exp(-(x_i-\theta))}{(1+\exp(-(x_i-\theta)))}.$$

Although we can not simplify this further one can show there is a unique solution to this equation.

Numerical maximisation of the likelihood: Newton-Raphson method

When the likelihood (or log-likelihood) cannot be maximised analytically (calculus, or considering graphs etc.), we can use numerical methods.

Here we consider the *Newton-Raphson algorithm*:

- It provides a simple way to find a root of an equation of the form $g(\theta) = 0$ where g is differentiable
- Requires an initial guess of the root, θ_0
- Use Taylor series to approximate g near θ_0

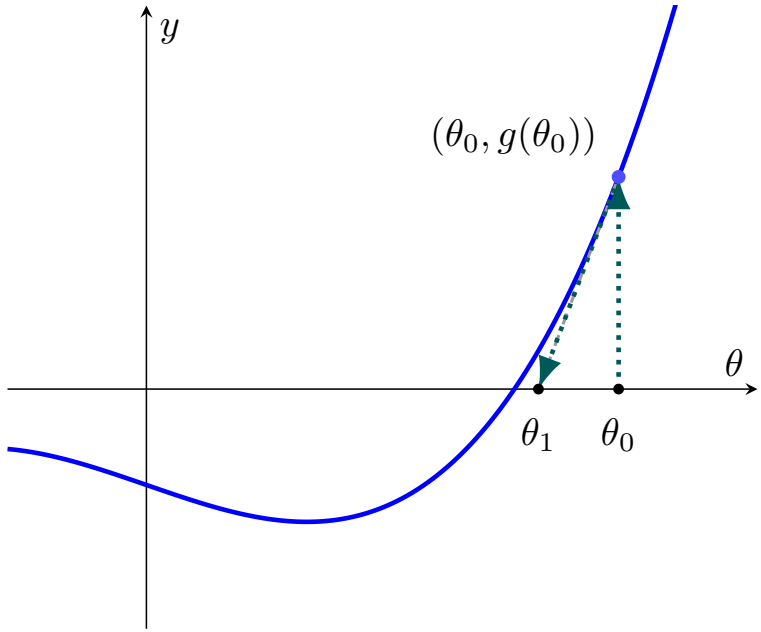
$$g(\theta) \approx g(\theta_0) + g'(\theta_0)(\theta - \theta_0)$$

- Find the root of the approximation

$$\theta_1 = \theta_0 - \frac{f(\theta_0)}{f'(\theta_0)}$$

- θ_1 is our new guess for the root of g.

Having an initial value θ_0, obtain a 'better' approximation of the root:



We have that

$$\theta_1 = \theta_0 - \frac{g(\theta_0)}{g'(\theta_0)}.$$

Now we can apply the same procedure to θ_1 to obtain an even better guess. In this way we can generate a sequence of estimates $\theta_1, \theta_2, \ldots, \theta_i, \ldots$ where

$$\theta_{i+1} = \theta_i - \frac{g(\theta_i)}{g'(\theta_i)}.$$

Logistic Distribution continued

The Newton-Raphson method will converge to the root very quickly provided that the initial value θ_0 is a reasonably good guess.

